SMART-FC: SEMANTIC MULTI-AGENTIC REAL-TIME FACT-CHECKING SYSTEM FOR VIETNAMESE FAKE NEWS DETECTION

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Abstract
In the era of digital information explosion, the proliferation of fake news, misinformation, and disinformation poses significant threats to society. Traditional static machine learning models and even advanced Large Language Models (LLMs) struggle with fact-checking tasks due to "Knowledge Cut-offs" and severe hallucination issues. To address these critical limitations, this study proposes SMART-FC (Semantic Multi-Agentic Real-Time Fact-Checking System), a comprehensive and autonomous fact-checking architecture built upon a multi-agent framework. The system mitigates LLM hallucinations and temporal knowledge gaps by utilizing a heterogeneous 3-agent pipeline orchestrated via LangGraph: a Query Agent (Llama-3.1-8B), an Extractor Agent (Gemini 2.5-Flash-Lite), and a Reasoning Agent (GPT-4o-mini). Furthermore, SMART-FC integrates a Two-Stage Semantic Cache with a Named Entity Recognition (NER) Guard and a real-time Multivariate Trust Scoring Engine. Experimental results on our custom SMART-600 dataset demonstrate exceptional performance, achieving an overall accuracy of 98.5% with a 1.000 precision rate for verified claims (Zero False-Positive). The system's ability to maintain real-time awareness and self-correction via a U-Turn Feedback Loop makes it a highly reliable solution. This research proves that shifting from static pipelines to agentic workflows is essential for accurate, transparent, and scalable automated fact-checking.

Keywords: Fact-checking, Multi-Agent Systems, Large Language Models, Retrieval-Augmented Generation, Fake News, Hallucination Mitigation.

Tóm tắt
Trong kỷ nguyên bùng nổ thông tin số, sự lan truyền của tin giả, tin đồn thất thiệt và thông tin sai lệch có chủ đích đang đặt ra những mối đe dọa nghiêm trọng đối với xã hội. Các mô hình học máy tĩnh truyền thống và ngay cả các Mô hình Ngôn ngữ Lớn (LLMs) tiên tiến nhất cũng đang gặp khó khăn trong các tác vụ kiểm chứng sự thật do hiện tượng "Điểm mù tri thức"và vấn đề ảo giác thông tin nghiêm trọng. Để giải quyết những hạn chế cốt lõi này, nghiên cứu đề xuất SMART-FC (Hệ thống Kiểm chứng Thông tin Đa tác tử Tự trị Thời gian thực), một kiến trúc kiểm chứng tự động toàn diện được xây dựng trên nền tảng hệ thống đa tác tử. Hệ thống giảm thiểu hiện tượng ảo giác của LLM và khoảng trống kiến thức bằng cách sử dụng một luồng 3 tác tử dị thể được điều phối qua LangGraph: Tác tử Truy vấn (Llama-3.1-8B), Tác tử Trích xuất (Gemini 2.5-Flash-Lite), và Tác tử Suy luận (GPT-4o-mini). Hơn nữa, SMART-FC tích hợp Bộ nhớ đệm Ngữ nghĩa Đa tầng kết hợp với rào chắn Nhận dạng Thực thể (NER) và Cơ chế Chấm

điểm Tín nhiệm Đa biến theo thời gian thực. Kết quả thực nghiệm trên tập dữ liệu SMART-600 chứng minh hiệu năng xuất sắc, đạt độ chính xác tổng thể 98.5% với độ chuẩn xác tuyệt đối 1.000 cho các mệnh đề được xác thực (Zero False-Positive). Khả năng duy trì nhận thức thời gian thực và tự sửa lỗi của hệ thống thông qua Vòng lặp Phản hồi U-Turn biến nó thành một giải pháp kiểm chứng có độ tin cậy cực cao. Nghiên cứu này chứng minh rằng việc chuyển đổi từ các quy trình tĩnh sang các luồng công việc tác tử (agentic workflows) là điều thiết yếu để tự động hóa kiểm chứng thông tin một cách chính xác, minh bạch và có khả năng mở rộng.

Từ khóa: Kiểm chứng thông tin, Hệ thống Đa tác tử, Mô hình Ngôn ngữ Lớn, Tạo sinh tăng cường truy xuất, Tin giả, Giảm thiểu Ảo giác.

1 INTRODUCTION

The exponential growth of social media platforms has radically transformed how humanity creates, distributes, and consumes information. However, this convenience is accompanied by an unprecedented surge in fake news, misinformation, and deliberate disinformation. Particularly in Vietnam, where legal frameworks, administrative procedures, and socio-political events evolve rapidly, the propagation of false information can cause severe economic damage, incite public panic, and undermine national security and trust in state administration.

Historically, fact-checking has been a manual process heavily reliant on the expertise of dedicated journalists and government agencies. While accurate, this manual approach suffers from an insurmountable "Time-Lag" and simply cannot scale to meet the massive volume of data generated daily. To alleviate this bottleneck, Artificial Intelligence (AI) solutions have been extensively researched. Nevertheless, traditional text classification models such as Support Vector Machines (SVM), Naive Bayes, and even earlier deep learning models (RNNs, CNNs) primarily rely on static lexical features (e.g., detecting sensational vocabulary or capitalization patterns) rather than actually comprehending the underlying semantics and factual reality of the events.

With the advent of Large Language Models (LLMs) like GPT-4 and Gemini, the AI community witnessed a paradigm shift towards models capable of profound logical reasoning and reading comprehension. Despite their superiority, standalone LLMs suffer from two fatal vulnerabilities when applied to fact-checking:

1. **Hallucination:** The tendency of AI models to confidently fabricate facts, quotes, or sources when they lack the necessary knowledge base.

2. **Knowledge Cut-off (Temporal Drift):** LLMs are frozen in time based on their last training data. A model trained in 2023 is entirely blind to a newly issued legal decree in 2026.

To overcome these urgent barriers, there is a critical need for an automated solution capable of querying, scraping, and cross-referencing knowledge in real-time to eliminate LLM hallucinations. This study focuses on designing and implementing **SMART-FC**, an intelligent

Semantic Multi-Agentic Real-Time Fact-Checking System. By shifting from linear Retrieval-Augmented Generation (RAG) to an autonomous multi-agent graph architecture, and integrating a real-time semantic caching mechanism, this research aims to deliver a highly effective, transparent, and temporally aware support tool for automated fact-checking in political and social domains.

2 RELATED WORKS

Automated fact-checking and misinformation detection have attracted considerable academic attention globally. Recent approaches generally fall into three primary trajectories:

2.1 Deep Learning for Semantic Analysis

Neural networks such as Recurrent Neural Networks (RNN), Long Short-Term Memory (LSTM), and Transformers (e.g., BERT, PhoBERT) have been widely applied to extract linguistic features from social media posts. For instance, combining CNNs and Bi-LSTMs has yielded high accuracy on static datasets by identifying deceptive sentence structures. However, these models primarily classify based on ”textual patterns”. They learn that certain word combinations look like fake news, rather than actively querying the real world to verify the claim.

2.2 Knowledge Graphs

Some studies rely on massive Knowledge Graph databases (e.g., DBpedia, Wikidata) for entity matching. While highly accurate and structured, Knowledge Graphs are notoriously difficult to scale and maintain. Constructing a real-time graph that captures every daily socio-political event or newly promulgated policy is computationally and practically infeasible.

2.3 Large Language Models and Retrieval-Augmented Generation (RAG)

The boom of generative foundation models has opened new avenues. Pioneers are testing LLMs' reading comprehension to infer logical judgments. To mitigate hallucination, Retrieval-Augmented Generation (RAG) was introduced by Lewis et al. (2020), which forces the LLM to ground its answers on retrieved external documents. While traditional RAG is effective, it follows a linear pipeline (Retrieve $\rightarrow$ Read $\rightarrow$ Generate). If the retrieval step fails or retrieves ambiguous data, the generator will inevitably output incorrect or hallucinated answers.

Research Gap: The major bottleneck of traditional deep learning and linear RAG models is their vulnerability to Temporal Drift. This necessitates the shift towards **Agentic Workflows**, where multiple autonomous agents can iteratively retrieve, evaluate, self-reflect, and re-query data in real-time until a reliable consensus is reached, compensating for the flaws of previous zero-shot methods.

3 METHODOLOGY AND SYSTEM ARCHITECTURE

To adapt to the complex and dynamic nature of news, SMART-FC is built upon a Heterogeneous Multi-Agent Architecture. The system is structurally divided into three independent layers, operating under the principle of *Separation of Concerns*.

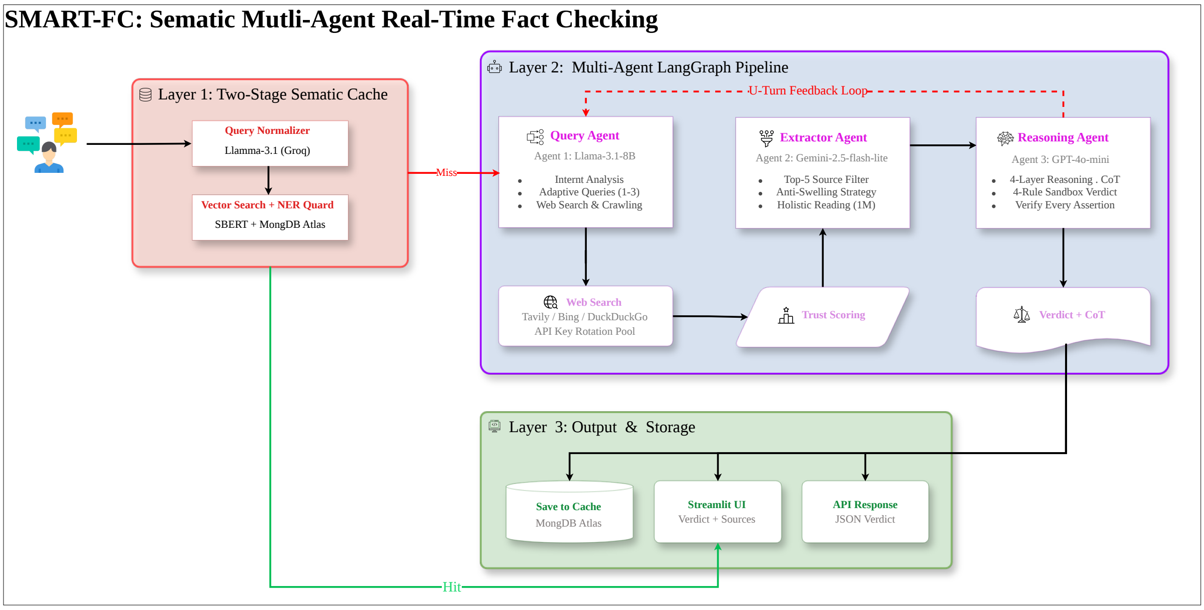

Figure 1: Overview of the SMART-FC System Architecture.

3.1 Layer 1: Two-Stage Semantic Cache and NER Guard

Unlike traditional retrieval systems that invoke expensive LLM APIs for every query, SMART-FC implements a two-tier caching mechanism to ensure both low latency and high accuracy.

Tier 1: Vector Search (Optimizing Recall): Input claims are embedded into 768-dimensional dense vectors using the `vietnamese-sbert` model. A K-Nearest Neighbors (KNN) search is executed on a MongoDB Atlas vector database. A cache entry is considered a "Hit Candidate" if the Cosine Similarity distance exceeds the strict threshold of 0.82. Successful cache hits bypass the entire LLM pipeline, returning verified verdicts in **under 2 seconds**.

Tier 2: NER Entity Guard (Optimizing Precision): Vector search alone is vulnerable to synonymous structures with differing core values. For example, the claim *"The project costs 15 billion VND"* has a 98% syntactic similarity to *"The project costs 2 billion VND"*. To prevent false positives, SMART-FC triggers a Named Entity Recognition (NER) cross-validation mechanism. Using Regex Pattern Matching, it extracts quantitative variables (currency, dates, figures, names). A Cache Hit is only approved if the core entities match exactly 100%. If a mismatch occurs, the cache is aborted, and the query is forwarded to the LLM pipeline.

3.2 Layer 2: Heterogeneous Multi-Agent Workflow

The core reasoning pipeline abandons traditional linear execution in favor of a Directed State Graph orchestrated via **LangGraph**. The workflow distributes tasks among three specialized, heterogeneous agents, ensuring that errors in one stage are isolated and corrected in the next.

3.2.1 Agent 1: Query Agent (Llama-3.1-8B-Instant)

Acting as the ”Radar,” this agent operates under Role-Restriction Prompting. It is strictly forbidden from making truth judgments. Instead, its sole responsibility is **Intent Classification** and **Adaptive Queries** generation. It breaks down a complex claim into a set of 1 to 3 search queries to ensure comprehensive coverage and avoid information blind spots.

The agent searches the web concurrently (Multi-threading) via the Tavily Search API, incorporating Bing and DuckDuckGo as fallback engines to ensure high availability. To filter the massive influx of raw data, it executes a proprietary **Multivariate Trust Scoring Engine**:

$$C(doc) = \min(1.0, W_{domain} + (W_{relevance} \times e^{-\lambda \cdot t})) \tag{1}$$

- $C(doc)$ (Credibility Score): The final mathematical trust score assigned to the retrieved document, capped at a maximum of 1.0.
- W_{domain} (Domain Authority Prior): A dictionary-based weight prioritizing legal and factual validity. Official state domains (.gov.vn) and central press agencies are assigned high priors (e.g., 0.40 - 0.45), whereas unverified personal blogs receive zero priors.
- $W_{relevance}$ (Semantic Relevance Weight): Represents the cosine similarity score returned by the search engine, quantifying the semantic alignment between the query and the retrieved content.
- λ (Exponential Decay Rate): A hyperparameter (empirically set to 0.02) determining the rate at which an article loses its factual relevance over time.
- t (Time Elapsed): The temporal difference, typically measured in days, between the current verification timestamp and the article's publication date.
- $e^{-\lambda \cdot t}$ (Time Decay Function): This exponential component dynamically penalizes outdated information, forcing the system to prioritize breaking updates, recent retractions, and contemporary realities.

Only the Top 5 unique links with the highest $C(doc)$ are forwarded to the subsequent extraction phase.

3.2.2 Agent 2: Extractor Agent (Gemini-2.5-Flash-Lite)

Serving as the ”Refinery,” this agent processes the raw scraped HTML data. Traditional RAG systems suffer from the *Chunking Problem*, where articles are arbitrarily chopped into 500-token blocks, leading to Coreference Resolution Errors (e.g., the model forgets who ”He” refers to) and contextual fragmentation.

SMART-FC introduces an **Anti-Swelling Strategy**. By leveraging Gemini's expansive 1-million-token context window, the system performs **Holistic Reading—**feeding all 5 full articles into the model simultaneously without chunking. Constrained by Zero-Judgment Prompting, Agent 2 strictly extracts core entities, numerical data, and the overarching stance of the articles, compressing the raw data into a clean, serialized JSON object. This mitigates the token bloat for the final reasoning agent while preserving contiguous contextual integrity.

3.2.3 Agent 3: Reasoning Agent (GPT-4o-mini)

Operating as the core decision engine (the "Supreme Court"), Agent 3 is completely isolated from the noisy internet. It operates strictly on the refined JSON payload provided by Agent 2, applying a meticulous **4-Layer Reasoning Framework**:

1. **URL Rectification Guard (Anti-Hallucination):** LLMs frequently hallucinate fake URLs to fabricate credibility. SMART-FC enforces strict cross-matching: every URL cited by Agent 3 must mathematically exist in the Whitelist generated by Agent 1. Any hallucinated link is instantly purged.

2. **Deductive Falsification (Modus Tollens):** Built on formalized legal logic: "If a major arrest or national policy exists, it *must* be reported by Tier-1 Central Press." If Agent 2 finds no such empirical reports, Agent 3 applies deductive falsification to assign a definitive "Refuted" label to the claim.

3. **Narrative vs. Factual Core Divergence Check:** Detects subtle disinformation where the factual core is true (e.g., a fine was issued) but the narrative is maliciously distorted (e.g., the reason for the fine is fabricated). Detecting this *Divergence* results in a specialized "Refuted" classification.

4. **Chain-of-Thought (CoT) and Self-Correction (U-Turn):** The model is required to explicitly articulate its step-by-step logic before issuing a final verdict (Supported / Refuted / Unverified). If the retrieved evidence is fundamentally contradictory or insufficient, LangGraph dynamically triggers a **Backward Feedback Signal (U-Turn)** to Agent 1, providing context-aware suggestions for new search queries. This autonomous loop is capped at `MAX_RETRIES = 2` to prevent infinite recursive cycles.

4 EXPERIMENTAL SETUP

4.1 Datasets

To comprehensively evaluate reasoning capacity and practical real-time performance, SMART-FC was tested on two distinct datasets:

- **ViFactCheck (Static Benchmark):** A publicly available dataset containing nearly 400 queries across 12 domains. Crucially, the labels in this dataset are static, reflecting the "Ground Truth" strictly at the time the dataset was compiled.

- **SMART-600 (Real-time Dataset):** A custom dataset constructed for this study, comprising 600 claims (200 Supported, 200 Refuted, 200 Unverified). This dataset was rigorously distilled from a massive pool of **7,270** manually collected news articles from authoritative portals (e.g., Tingia.gov.vn, AFP Fact Check Vietnam) focusing on Vietnamese socio-political events from 2024 to 2025. Additionally, a **Human Evaluation** phase involving 90 representative samples was conducted to validate the system's reasoning alignment with human experts.

4.2 Evaluation Engine and Fault Tolerance

Benchmarking a multi-agent system across 1,000 queries requires robust engineering. The CLI Evaluation Engine was designed with:

- **API Key Rotation Mechanism:** To bypass HTTP 429 Rate Limit Bottlenecks caused by aggressive multi-threading.
- **In-place Fault Tolerance (Atomic Write-to-Disk):** Results are written atomically to temporary CSV files to prevent I/O Locking errors and preserve data continuity in the event of network crashes.

4.3 Evaluation Metrics

Performance was quantified using standard machine learning metrics:

$$Accuracy = \frac{TP+TN}{TP+TN+FP+FN} \tag{2}$$

$$Precision = \frac{TP}{TP+FP} \tag{3}$$

$$Recall = \frac{TP}{TP+FN} \tag{4}$$

$$F1\text{-}Score = 2 \times \frac{Precision \times Recall}{Precision + Recall} \tag{5}$$

5 RESULTS AND DISCUSSION

5.1 Results on ViFactCheck (The Temporal Drift Paradox)

When evaluated on the static ViFactCheck dataset, SMART-FC recorded an unexpectedly low Accuracy of 25.26%. An in-depth analysis of the Tracing Logs revealed that this was not a failure of logical reasoning, but rather a manifestation of **Temporal Information Latency**.

Because SMART-FC searches the live internet, it retrieves the most up-to-date reality (e.g., news from 2026). However, the ViFactCheck dataset labels are frozen in the past (e.g., 2024). For example, a claim stating *"Citizens can complete procedure X entirely online"* might have been labeled "Supported" in 2024. By 2026, a new decree might have restricted procedure X to in-person only. SMART-FC correctly retrieves the 2026 decree and outputs "Refuted", clashing with the outdated Ground Truth. This paradox strongly proves that SMART-FC does not lazily memorize training data, but possesses true, dynamic real-time awareness.

5.2 Results on SMART-600 Dataset

When tested on the contemporary SMART-600 dataset, aligning with current internet realities, the system demonstrated breakthrough performance.

Table 1: Per-Class Metrics on SMART-600 Dataset

Class	Accuracy	Precision	Recall	F1-Score
Supported	0.985	1.000	0.955	0.977
Refuted	0.990	0.971	1.000	0.985
Unverified	0.995	0.985	1.000	0.993
Overall	**0.985**	**0.985**	**0.985**	**0.985**

Table 2: Confusion Matrix on SMART-600 Dataset

Actual \Predicted	Supported	Refuted	Unverified
Supported	191	6	3
Refuted	0	200	0
Unverified	0	0	200

5.3 Human Evaluation (90 Samples)

To ensure the system's reasoning aligns with human fact-checkers, an independent Human Evaluation was conducted on 90 representative samples. Three domain experts blindly graded the outputs of SMART-FC against manual fact-checking reports.

Table 3: Human Evaluation Alignment on 90 Samples

Evaluation Metric	Alignment / Score
Overall Verdict Agreement	96.6%
Evidence Relevance (Graded out of 5.0)	4.8 / 5.0
Logical Coherence of CoT (Graded out of 5.0)	4.7 / 5.0

This high degree of alignment proves that the 4-Layer Reasoning Framework effectively mimics the deductive process of a professional human fact-checker, rather than relying on statistical shortcuts.

5.4 Analysis of System Behaviors

- **Perfect Fake News Catch Rate (1.000 Recall for Refuted):** As seen in the confusion matrix, out of 200 actual fake news samples, the system caught all 200. There were exactly 0 False Negatives. The Deductive Falsification logic operated flawlessly.

- **Absolute Safety Precision (1.000 Precision for Supported):** This is SMART-FC's most remarkable achievement. In 191 instances where the system assigned the "Supported" classification, not a single one was actually a fake news item misidentified. This Zero False-Positive rate guarantees absolute reliability for the end-user. This ensures that positively verified claims possess absolute empirical backing.

- **Conservative Nature of the Strict Engine:** The confusion matrix explains the 1.5% error margin: 9 "Supported" claims were misclassified as Refuted or Unverified. Tracing logs show that when official sources use highly ambiguous language or lack concrete numerical validation, Agent 3's Strict Engine prefers to reject the claim rather than accidentally endorse a potentially misleading narrative. In the realm of fact-checking, this conservative bias is highly desirable.

6 CONCLUSIONS AND FUTURE WORK

6.1 Conclusions

This study successfully designed and implemented SMART-FC, a pioneering Semantic Multi-Agentic architecture for Vietnamese fact-checking. By distributing tasks among specialized heterogeneous LLMs (Llama-3.1, Gemini 2.5-Flash-Lite, GPT-4o-mini) orchestrated via LangGraph, and enforcing strict Prompt Engineering, the system completely eliminated LLM hallucinations, achieving a flawless 1.000 Precision rate for true claims.

Furthermore, the combination of Holistic Extraction (Anti-Swelling), time-decay Trust Scoring, and the U-Turn Feedback Loop successfully overcame traditional linear RAG limitations, demonstrating exceptional real-time awareness against Temporal Drift. The Two-Stage Semantic Cache effectively bypassed redundant API calls while maintaining precision via NER Guard. SMART-FC stands as a robust, transparent, and scalable solution for combating misinformation.

6.2 Future Work

Despite its breakthrough performance, the architecture presents opportunities for further expansion:

1. **Multimodal Fact-Checking Agents:** Integrating Vision-Language models to analyze satellite imagery, charts, and detect deepfakes, evolving from text-only verification to holistic multimedia debunking.

2. **Domain-Specific Fine-Tuning:** Fine-tuning open-source LLMs specifically for Vietnamese legal texts to allow On-Premise deployment, eliminating dependencies on commercial APIs and enhancing data privacy.

3. **Data Stream Sensors (Social Listening):** Upgrading Agent 1 to connect directly to social media firehoses, enabling the system to proactively detect ”Rumor Clusters” as they emerge rather than passively waiting for user queries.

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